

Example 25. Solve $\frac{dy}{dx} = \frac{2x + 3y + 4}{4x + 6y + 5}$.

Solution. Let $2x + 3y = v$.

Then $2 + 3 \frac{dy}{dx} = \frac{dv}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{3} \left(\frac{dv}{dx} - 2 \right)$.

Using above substitutions, the given differential equation reduces to

$$\frac{1}{3} \left(\frac{dv}{dx} - 2 \right) = \frac{v + 4}{2v + 5}$$

$$\frac{dv}{dx} - 2 = \frac{3v + 12}{2v + 5}$$

or

$$\frac{dv}{dx} = 2 + \frac{3v + 12}{2v + 5} = \frac{7v + 22}{2v + 5}$$

or

Separating the variables, we get

$$\frac{2v + 5}{7v + 22} dv = dx.$$

Integrating both sides, we get

$$\int \frac{2v + 5}{7v + 22} dv = \int dx + C$$

$$\Rightarrow 2 \int \frac{v}{7v + 22} dv + 5 \int \frac{dv}{7v + 22} = x + C$$

$$\Rightarrow \frac{2}{7} \int \frac{7v}{7v + 22} dv + 5 \int \frac{dv}{7v + 22} = x + C$$

$$\Rightarrow \frac{2}{7} \int \frac{7v + 22 - 22}{7v + 22} dv + 5 \int \frac{dv}{7v + 22} = x + C$$

$$\Rightarrow \frac{2}{7} \int dv - \frac{44}{7} \int \frac{dv}{7v + 22} + 5 \int \frac{dv}{7v + 22} = x + C$$

$$\Rightarrow \frac{2}{7} v - \frac{9}{7} \int \frac{dv}{7v + 22} = x + C$$

$$\Rightarrow \frac{2}{7} v - \frac{9}{49} \log | 7v + 22 | = x + C$$

$$\Rightarrow 14v - 9 \log | 7v + 22 | = 49x + 49C$$

$$\Rightarrow 14(2x + 3y) - 9 \log | 14x - 21y + 22 | = 49x + 49C$$

$$\Rightarrow 9 \log | 14x + 21y + 22 | = 42y - 21x + k, k = 49C$$

which is the required solution.

Example 3. Solve $(1 + x^2) \frac{dy}{dx} + y = \tan^{-1} x$.

Solution. The given differential equation can be written as

$$\frac{dy}{dx} + \frac{1}{1+x^2} y = \frac{\tan^{-1} x}{1+x^2}.$$

Comparing (1) with equation $\frac{dy}{dx} + Py = Q$, we get

$$P = \frac{1}{1+x^2} \text{ and } Q = \frac{\tan^{-1} x}{1+x^2}$$

so,

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{dx}{1+x^2}} = e^{\tan^{-1} x}.$$

Thus, the solution of (1) is

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dx + C$$

$$\Rightarrow y \cdot (e^{\tan^{-1} x}) = \int \frac{\tan^{-1} x}{1+x^2} e^{\tan^{-1} x} dx + C$$

$$= \int t e^t dt + C, \text{ where } t = \tan^{-1} x$$

$$= t \int e^t dt - \int \left\{ \frac{d}{dt}(t) \cdot \int e^t dt \right\} dt + C \quad [\text{Integrating by parts}]$$

$$= te^t - \int e^t dt + C = te^t - e^t + C$$

$$= (\tan^{-1} x - 1) e^{\tan^{-1} x} + C.$$

$$\therefore y = (\tan^{-1} x - 1) + Ce^{-\tan^{-1} x}$$

which is the required solution.

Example 4. Solve $(D^2 - 4D + 4) y = x^2 + e^x + \cos 2x$.

(Meerut 2000, 2001)

Solution. Here, the given differential equation is

$$(D^2 - 4D + 4) y = x^2 + e^x + \cos 2x. \quad \dots(1)$$

To find C.F., the auxiliary equation is given by

$$m^2 - 4m + 4 = 0$$

$$\Rightarrow (m - 2)^2 = 0 \Rightarrow m = 2, 2$$

$$\therefore \text{C.F.} = (C_1 + C_2 x) e^{2x}$$

$$\text{Now, P.I.} = \frac{1}{(D^2 - 4D + 4)} (x^2 + e^x + \cos 2x)$$

$$= \frac{1}{(D - 2)^2} x^2 + \frac{1}{(D - 2)^2} e^x + \frac{1}{(D^2 - 4D + 4)} \cos 2x$$

$$\begin{aligned} &= \frac{1}{4 \left(1 + \frac{D^2}{4} - D \right)} (x^2 + e^x + \cos 2x) \\ &= \frac{1}{4 \left(\left(\frac{D}{2} \right)^2 - 1 \right)^2} \end{aligned}$$

$$= \frac{1}{4 \left(1 - \frac{D}{2}\right)^2} x^2 + \frac{1}{(1-2)^2} e^x + \frac{1}{(-2^2 - 4D + 4)} \cos 2x$$

$$= \frac{1}{4} \left(1 - \frac{D}{2}\right)^{-2} x^2 + \frac{e^x}{1} - \frac{1}{4D} \cos 2x$$

$$= \frac{1}{4} \left(1 + D + \frac{3}{4} D^2 + \dots\right) x^2 + e^x - \frac{1}{4} \int \cos 2x \, dx$$

$$= \frac{1}{4} \left(x^2 + D(x^2) + \frac{3}{4} D^2(x^2)\right) + e^x - \frac{1}{4} \cdot \frac{1}{2} \sin 2x$$

$$= \frac{1}{4} \left[x^2 + 2x + \frac{3}{2}\right] + e^x - \frac{1}{8} \sin 2x$$

Hence, the complete solution is given by

$$y = \text{C.F.} + \text{P.I.}$$

$$\Rightarrow y = (C_1 + C_2 x) e^{2x} + \frac{1}{4} \left(x^2 + 2x + \frac{3}{2}\right) + e^x - \frac{1}{8} \sin 2x$$

Example 5. Solve $x^3 \frac{d^3 y}{dx^3} + 3x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = x + \log x$.

Solution. Putting $x = e^z$, in the given equation, we get

$$(D(D-1)(D-2) + 3D(D-1) + (D+1))y = e^z + z \text{ where } D = \frac{d}{dz}$$

$$\Rightarrow (D^3 + 1)y = e^z + z. \quad \dots(1)$$

To find the C.F. of (1), auxiliary equation is

$$m^3 + 1 = 0 \Rightarrow (m+1)(m^2 - m + 1) = 0$$

$$\Rightarrow m = -1, \frac{1 \pm i\sqrt{3}}{2}$$

$$\begin{aligned} \therefore \text{C.F.} &= C_1 e^{-z} + e^{z/2} \left[C_2 \cos\left(\frac{\sqrt{3}}{2}z\right) + C_3 \sin\left(\frac{\sqrt{3}}{2}z\right) \right] \\ &= C_1 x^{-1} + x^{1/2} \left[C_2 \cos\left(\frac{\sqrt{3}}{2} \log x\right) + C_3 \sin\left(\frac{\sqrt{3}}{2} \log x\right) \right] \end{aligned}$$

$$\begin{aligned} \text{Now, P.I.} &= \frac{1}{D^3 + 1} e^z + z = \frac{1}{D^3 + 1} e^z + \frac{1}{D^3 + 1} z \\ &= \frac{e^z}{1^3 + 1} + (1 + D^3)^{-1} z = \frac{1}{2} e^z + (1 - D^3 + \dots) z \\ &= \frac{1}{2} e^z + z = \frac{1}{2} x + \log x. \end{aligned}$$

Hence, the general solution of (1), is given by

$$y = \text{C.F.} + \text{P.I.}$$

$$\Rightarrow y = \frac{C_1}{x} + \sqrt{x} \left[C_2 \cos\left(\frac{\sqrt{3}}{2} \log x\right) + C_3 \sin\left(\frac{\sqrt{3}}{2} \log x\right) \right] + \frac{1}{2} x + \log x.$$

Example 4. Transform the differential equation

$$\cos x \frac{d^2 y}{dx^2} + \sin x \frac{dy}{dx} - 2y \cos^3 x = 2 \cos^5 x$$

into one having z as independent variable, where $z = \sin x$ and solve it.

(Meerut 2000)

Solution. Given that

$$z = \sin x \Rightarrow \frac{dz}{dx} = \cos x$$

Now, $\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \cos x \frac{dy}{dz}$

and $\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dz} \right) = \frac{d}{dx} \left(\cos x \frac{dy}{dz} \right) = -\sin x \frac{dy}{dz} + \cos x \frac{d}{dx} \left(\frac{dy}{dz} \right)$

$$= -\sin x \frac{dy}{dz} + \cos x \frac{d}{dz} \left(\frac{dy}{dz} \right) \cdot \frac{dz}{dx}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -\sin x \frac{dy}{dz} + \cos^2 x \frac{d^2 y}{dz^2} \quad [\text{By using (1)}]$$

Now, using (2) and (3), given equation reduces to

$$\cos x \left[-\sin x \frac{dy}{dz} + \cos^2 x \frac{d^2 y}{dz^2} \right] + \sin x \cos x \frac{dy}{dz} - 2 \cos^3 x \cdot y = 2 \cos^5 x$$

$$\Rightarrow \frac{d^2 y}{dz^2} - 2y = 2 \cos^2 x \Rightarrow \frac{d^2 y}{dz^2} - 2y = 2(1 - \sin^2 x)$$

$$\Rightarrow (D_1^2 - 2)y = 2(1 - z^2) \quad \left(D_1 = \frac{d}{dz} \right)$$

Now, the auxiliary equation is

$$m^2 - 2 = 0 \Rightarrow m = \pm \sqrt{2}$$

$$\therefore \text{C.F.} = C_1 e^{\sqrt{2}z} + C_2 e^{-\sqrt{2}z}$$

$$\text{P.I.} = 2 \cdot \frac{1}{D_1^2 - 2} (1 - z^2) = 2 \cdot \frac{1}{-2 \left(1 - \frac{D_1^2}{2} \right)} (1 - z^2)$$

$$= - \left(1 - \frac{D_1^2}{2} \right)^{-1} (1 - z^2) = - \left(1 + \frac{D_1^2}{2} + \dots \right) (1 - z^2) = - (1 - z^2 - 1) = z^2.$$

Therefore, the complete solution is given by

$$y = C_1 e^{z\sqrt{2}} + C_2 e^{-z\sqrt{2}} + z^2.$$

Putting $z = \sin x$, we get the required general solution as

$$y = C_1 e^{\sqrt{2} \sin x} + C_2 e^{-\sqrt{2} \sin x} + \sin^2 x.$$

Solve

$$1) \quad pz = 1 + q^2$$

Ans) $f(z, p, q) = 0$

Substitute $u = x + ay$

$\forall a = \text{arbitrary constant}$

Thus, $p = \frac{dz}{du}$ and $q = a \frac{dz}{du}$

Substitute this in our original eq -

$$z \left(\frac{dz}{du} \right) = 1 + \left(a \frac{dz}{du} \right)^2$$

$$z \frac{dz}{du} = 1 + a^2 \left(\frac{dz}{du} \right)^2$$

$$\Rightarrow a^2 \left(\frac{dz}{du} \right)^2 - z \frac{dz}{du} + 1 = 0$$

Now,

$$\frac{dz}{du} = -(-z) \pm \sqrt{(-z)^2 - 4a^2 \cdot 1}$$

$$\frac{dz}{du} = \frac{z \pm \sqrt{z^2 - 4a^2}}{2a^2}$$

$$du = \frac{2a^2}{z \pm \sqrt{z^2 - 4a^2}} dz$$

$$du = \frac{2a^2 (z \mp \sqrt{z^2 - 4a^2})}{(z \pm \sqrt{z^2 - 4a^2})(z \mp \sqrt{z^2 - 4a^2})}$$

After solving, we get

$$\frac{dz}{du} = \frac{1}{2} (z \mp \sqrt{z^2 - 4a^2})$$

Upon integrating on both sides, we get

$$\int du = \frac{1}{2} \int z dz + \frac{1}{2} \int \sqrt{z^2 - 4a^2} dz$$

$$u = \frac{z^2}{4} + \left[\frac{z \sqrt{z^2 - 4a^2}}{4} - \frac{(2a)^2}{2} \log(z + \sqrt{z^2 - 4a^2}) \right] + C$$

$$u = \frac{z^2}{4} + \frac{z}{4} \sqrt{z^2 - 4a^2} \pm a^2 \log|z + \sqrt{z^2 - 4a^2}| + C$$

Now,

After replacing this u with our original substitution, we get

$$x + ay = \frac{z^2}{4} + \frac{z}{4} \sqrt{z^2 - 4a^2} \pm a^2 \log(z + \sqrt{z^2 - 4a^2}) + C$$

where a and c are constants.

$$2) \quad 9(p^2 z + q^2) = 4$$

Here, $f(z, p, q) = 0$,
it belongs to a standard Form 3,

Substitute $u = z + a^2$, which gives us

$$\Rightarrow p = \frac{dz}{du} \quad q = a \frac{dz}{du}$$

Substituting this p and q in original eq.

$$9 \left(\frac{dz}{du} \right)^2 (z + a^2) = 4$$

$$\left(\frac{dz}{du} \right)^2 = \frac{4}{9(z + a^2)}$$

$$\frac{dz}{du} = \frac{2}{3\sqrt{z + a^2}}$$

$$3\sqrt{z+a^2} dz = 2du$$

$$3(z+a^2)^{1/2} dz = 2du$$

Integrating on both sides,

$$\int 3(z+a^2)^{1/2} dz = \int 2du$$

$$3 \left[\frac{(z+a^2)^{3/2}}{3/2} \right] = 2u + C$$

$$2(z+a^2)^{3/2} = 2u + C$$

Divide entire eq by 2, and $\frac{C}{2}$ is still C.

$$(z+a^2)^{3/2} = u + C$$

Replacing u with $u = x + ay$, we get

$$(z+a^2)^3 = (x+ay+C)^2$$

Example 9. Find the general solution of the following differential equation

$$(mz - ny) p + (nx - lz) q = ly - mx.$$

(Garhwal 1998, 2001; Meerut 1982, 86, 87, 88, 2006 B.P., 2008, 2009 B.P.)

Solution. Compare the given differential equation with Lagrange's differential equation $Pp + Qq = R$, we get

$$P = mz - ny, Q = nx - lz, R = ly - mx.$$

Then Lagrange's auxiliary equations are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx} \quad \dots(1)$$

Taking the multipliers x, y, z , then (1) becomes

$$\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx} = \frac{xdx + ydy + zdz}{0}$$

$$\therefore xdx + ydy + zdz = 0.$$

Integrating, we get

$$x^2 + y^2 + z^2 = c_1.$$

Again taking the multipliers l, m, n , then (1) becomes

$$\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx} = \frac{ldx + mdy + ndz}{0}$$

$$\therefore ldx + mdy + ndz = 0$$

Integrating we get

$$lx + my + nz = c_2.$$

Thus the general solution is

$$f(x^2 + y^2 + z^2, lx + my + nz) = 0.$$

Example 6. Show that $L \left\{ \frac{\sin t}{t} \right\} = \tan^{-1} \frac{1}{p}$ and hence find $L \left\{ \frac{\sin at}{t} \right\}$.

Does the Laplace transform of $\frac{\cos at}{t}$ exists?

(Rohilkhand 2002; Agra 1983, 85; Meerut 1991, 2011)

Solution. Let $F(t) = \sin t$

Then
$$\lim_{t \rightarrow 0} \frac{F(t)}{t} = \lim_{t \rightarrow 0} \frac{\sin t}{t} = 1.$$

We know that

$$L \{ \sin t \} = \frac{1}{p^2 + 1} = f(p) \text{ (say)}$$

Then, we have

$$\begin{aligned} L \left\{ \frac{\sin t}{t} \right\} &= \int_p^\infty f(x) dx = \int_p^\infty \frac{dx}{x^2 + 1} = \left(\tan^{-1} x \right)_p^\infty \\ &= \frac{\pi}{2} - \tan^{-1} p = \cot^{-1} p = \tan^{-1} \left(\frac{1}{p} \right). \end{aligned}$$

Now,
$$\begin{aligned} L \left\{ \frac{\sin at}{t} \right\} &= a L \left\{ \frac{\sin at}{at} \right\} \\ &= a \cdot \frac{1}{a} \tan^{-1} \left(\frac{1}{p/a} \right) \quad \left[\because L \{ f(at) \} = \frac{1}{a} f \left(\frac{p}{a} \right) \right] \\ &= \tan^{-1} \left(\frac{a}{p} \right). \end{aligned}$$

Also, since $L \{ \cos at \} = \frac{p}{p^2 + a^2} = f(p) \text{ (say)}$

Then
$$\begin{aligned} L \left\{ \frac{\cos at}{t} \right\} &= \int_p^\infty \frac{x}{x^2 + a^2} dx = \left[\frac{1}{2} \log (x^2 + a^2) \right]_p^\infty \\ &= \frac{1}{2} \lim_{x \rightarrow \infty} \log (x^2 + a^2) - \frac{1}{2} \log (p^2 + a^2) \end{aligned}$$

which does not exist, since $\lim_{x \rightarrow \infty} \log (x^2 + a^2)$ is infinite.

Therefore, $L \left\{ \frac{\cos at}{t} \right\}$ does not exist.

Example 5. Evaluate

$$(i) \quad L^{-1} \left\{ \frac{3p-2}{p^2-4p+20} \right\}$$

(Meerut 1987, 2011)

$$(ii) \quad L^{-1} \left\{ \frac{3p+7}{p^2-2p+3} \right\}.$$

(Meerut 1991P)

Solution. (i) We have

$$\begin{aligned} L^{-1} \left\{ \frac{3p-2}{p^2-4p+20} \right\} &= L^{-1} \left\{ \frac{3(p-2)+4}{(p-2)^2+16} \right\} \\ &= L^{-1} \left\{ \frac{3(p-2)}{(p-2)^2+16} + \frac{4}{(p-2)^2+16} \right\} \\ &= 3L^{-1} \left\{ \frac{p-2}{(p-2)^2+4^2} \right\} + 4L^{-1} \left\{ \frac{1}{(p-2)^2+4^2} \right\} \\ &= 3e^{2t} L^{-1} \left\{ \frac{p}{p^2+4^2} \right\} + 4e^{2t} L^{-1} \left\{ \frac{1}{p^2+4^2} \right\} \\ &= 3e^{2t} \cos 4t + e^{2t} \sin 4t. \end{aligned}$$

$$\begin{aligned} (ii) \quad L^{-1} \left\{ \frac{3p+7}{p^2-2p-3} \right\} &= L^{-1} \left\{ \frac{3(p-1)+10}{(p-1)^2-4} \right\} \\ &= L^{-1} \left\{ \frac{3(p-1)}{(p-1)^2-4} + \frac{10}{(p-1)^2-4} \right\} \end{aligned}$$

$$= 3L^{-1} \left\{ \frac{p-1}{(p-1)^2 - 4} \right\} + 10L^{-1} \left\{ \frac{1}{(p-1)^2 - 4} \right\}$$

$$= 3e^t L^{-1} \left\{ \frac{p}{p^2 - 2^2} \right\} + 10e^t L^{-1} \left\{ \frac{1}{p^2 - 2^2} \right\}$$

$$= 3e^t \cosh 2t + 5e^t \sinh 2t = 4e^{3t} - e^{-t}.$$